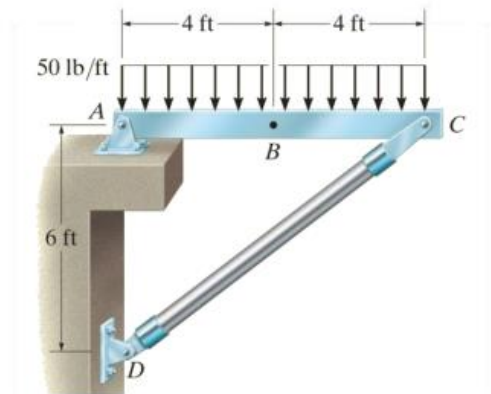


Exam 3 Practice

Practice Exam 3 – CE 2200, Dr. Koen

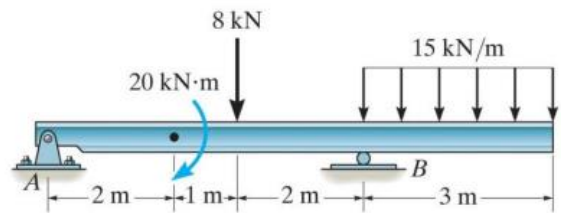
Section 1 – Internal forces at a point and equations.

- 1.1 Determine the normal force in the beam at point B.
- 1.2 Determine the shear force in the beam at point B.
- 1.3 Determine the bending moment in the beam at point B.
- 1.4 What is the shear force at any point along member DC?



- 1.5 Write the shear equation for the segment between $5 < x < 8$ (origin of x-axis is at point A).

Support Reactions $A_y = 14.3 \text{ kN} \downarrow$ and $B_y = 67.3 \text{ kN} \uparrow$



- 1.6 Write the moment equation for the segment between $5 < x < 8$ (origin of x-axis is at point A)

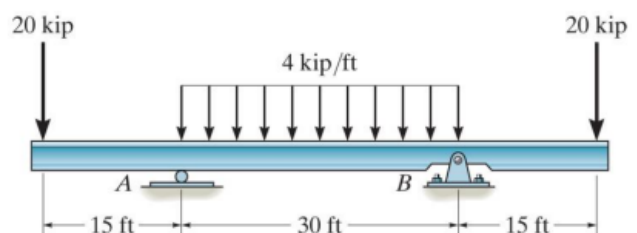
Support Reactions $A_y = 14.3 \text{ kN} \downarrow$ $B_y = 67.3 \text{ kN} \uparrow$

Section 2 – Graphing VM Diagrams with 6 rules

Support reactions.....

$A_y = B_y = 80 \text{ kip} \uparrow$

The x axis has its origin at the left side of the beam



- 2.1 What is the numerical value of change (magnitude) of the **change in shear** for the section $15 < x < 45 \text{ ft}$ and what is the **line type** for the connection?

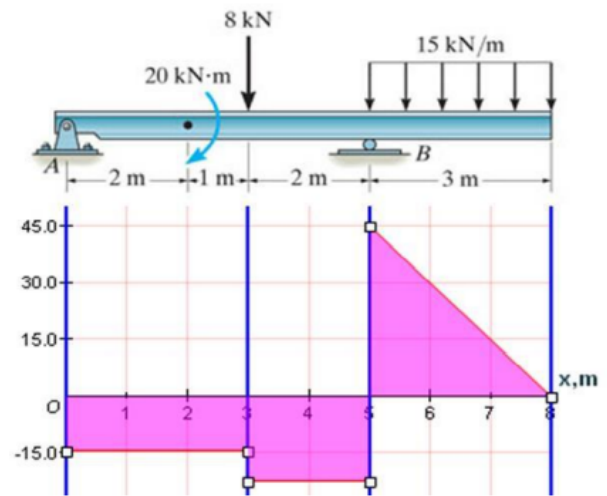
_____ kip — / \ \ \ / /

- 2.2 What happens to the **shear** diagram at $x = 45 \text{ ft}$?

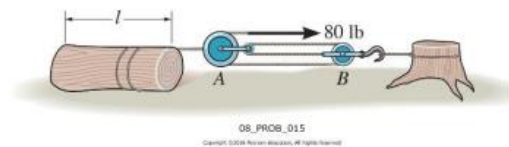
The x axis has its origin at the left side of the beam

_____ kN-m — / \) () (

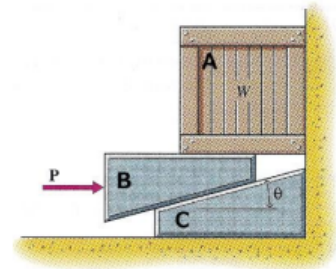
2.5 What happens to the moment diagram at $x=8$?



3.1 (5 pt) The log has a coefficient of static friction of $\mu = 0.3$ with the ground and a weight of 40 lb/ft. If a man can pull on the rope with a maximum force of 80 lb, determine the greatest length " L " of the log he can bring to the point of impending motion.



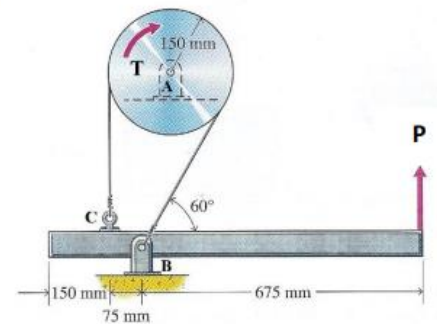
3.3 (2 pt) For Block A, determine the direction of the friction force acting on the bottom of Block A as a result of the impending motion of Wedge B



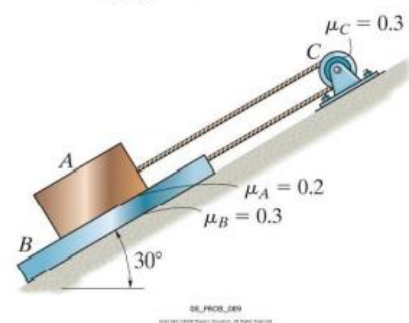
When calculating the minimum force P required to prevent motion of the wheel....

3.4 (4 pt) What is the angle beta to be used in the equation for belt friction when working with this system?

3.5 (4 pt) Which tension would be T_2



3.6 (6 pt) When determining the smallest mass of block A that would **prevent motion**, which tension would be T_2 and what would Beta equal?



Section 4 – Centroids and theorems of P&G

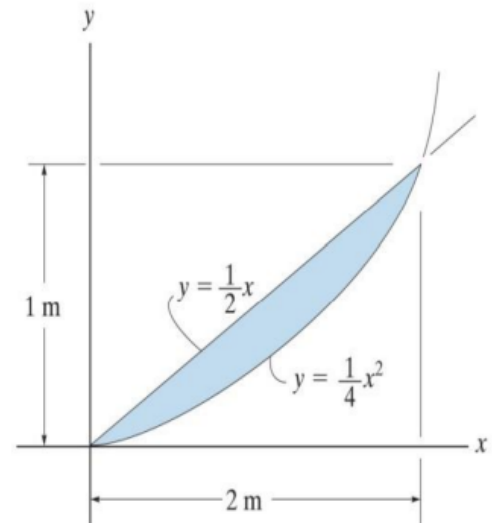
10 (pts) When finding the centroid by single integration with using a vertical element, which are the correct expressions?

4.1 $\tilde{x} =$

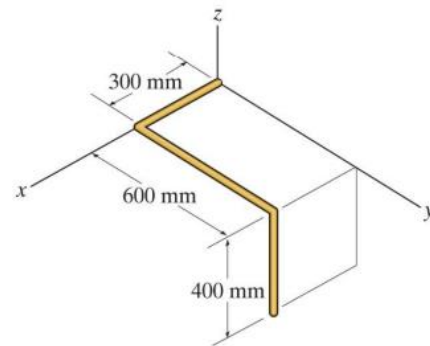
4.2 $\tilde{y} =$

4.3 $dA =$

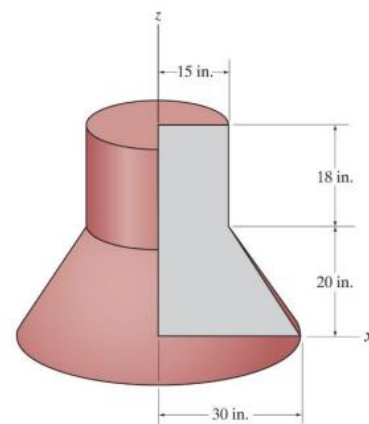
4.4 What are the limits for integration?



4.5 (5 pt) Determine the location of the centroid $\bar{y}$ -bar for the bent wire shape.



4.6 (10 pt) Determine the outside (**not** including the bottom) surface area and volume of the solid formed by a full rotation of the shaded area.



Answers

1.1 + 267 lb

1.2 0 lb

1.3 + 400 lb-ft

1.4 0 lb

1.5 $\{120 - 15x\}$ kN

1.6 $\{-480 + 120x - 7.5x^2\}$ kN-m

2.1 120 kip 

2.2 jumps up 80 kip

2.3 67.5 kN-m 

2.4 jumps 20 kN-m

2.5 no change

3.1 26.7 ft

3.2 221 N

3.3 to the right

3.4 3.665 rad

3.5 cable connected at C

3.6 cable connected to block B pi radians

4.1 x

4.2 $(.5x + .25x^2)/2$

4.3 $(.5x - .25x^2) dx$

4.4 0 to 2

4.5 323 mm

4.6 surface area = 5938 in² vol = 45,710 in³